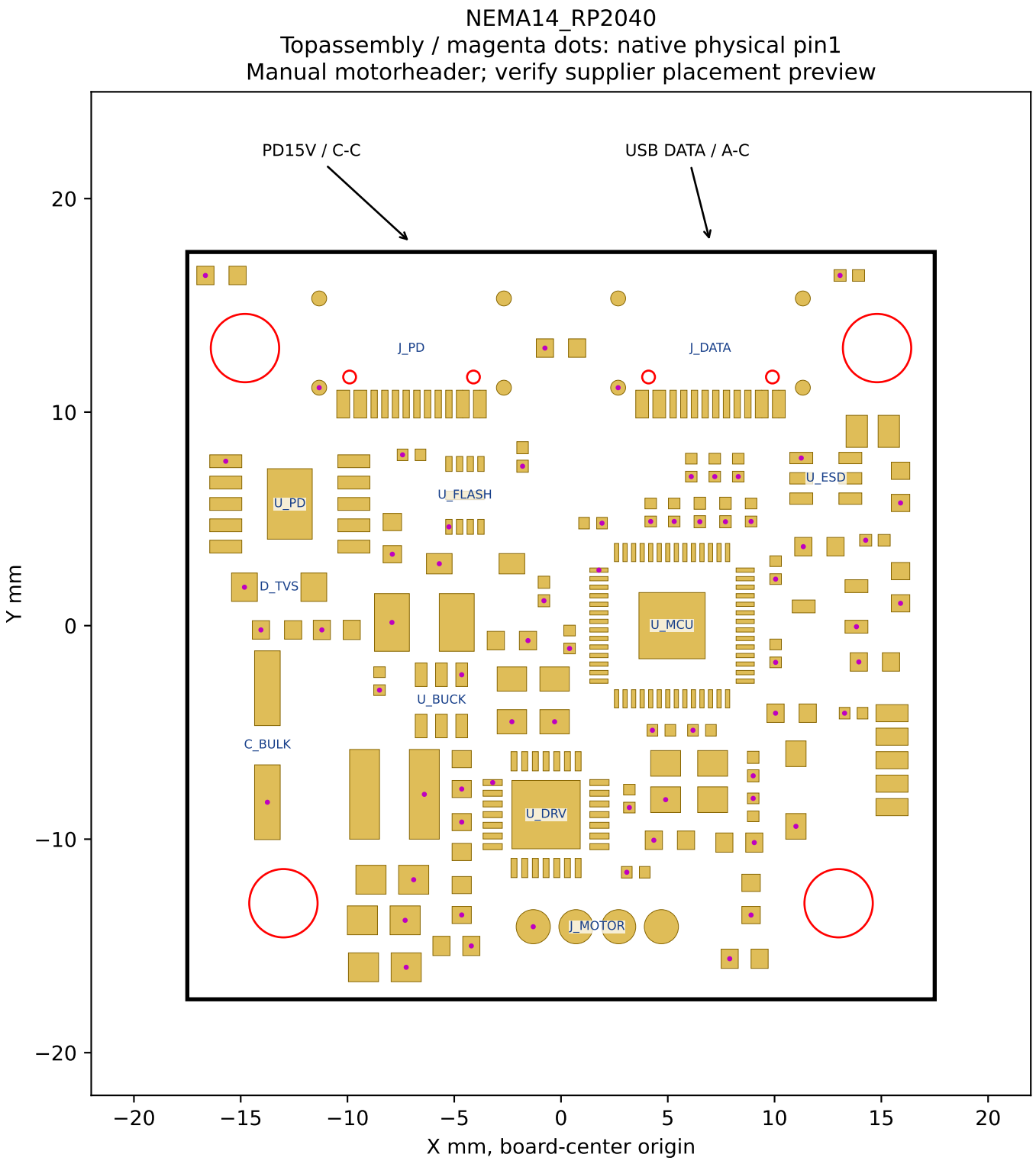


Motor: 1 BlackA+ / 2 GreenA- / 3 RedB+ / 4 BlueB-
PCB NPTH lower26mm / upper29.6mm pitch; front motor26mm pattern is separate.
Cap+ and every diode/IC orientation must match native pads/netlist and supplierpreview.
Filled/capped via list: via_process.csv. See fabrication-notes and mechanical-fit.
SHA256 0d5c64228122dc31dace9bb30ffdf36574f049d332c72365e2a2fd1528a2bc6a

NEMA14_RP2040 - fitted placement reference
Native source convention; supplier pickup/rotation approval pending

Ref	Exact MPN	Native center mm	Rotation	JLCPCB
J_PD	TYPE-C-31-M-12	-7.000,12.975	180	C165948
J_DATA	TYPE-C-31-M-12	7.000,12.975	180	C165948
U_PD	CH224K	-12.700,5.700	270	C970725
R_PD	ERJPA3F1001V	-13.300,-0.200	0	C441922
C_PD	CL10A105KB8NNNC	-10.500,-0.200	0	C15849
R_PG	0603WAF1002T5E	-7.900,4.100	90	C25804
U_MCU	RP2040	5.200,0.000	0	C2040
C_IO1	CL05B104KO5NNNC	1.500,4.800	180	C1525
C_IO10	CL05B104KO5NNNC	0.400,-0.650	90	C1525
C_IO22	CL05B104KO5NNNC	4.700,-4.900	0	C1525
C_IO33	CL05B104KO5NNNC	10.050,-1.300	90	C1525
C_IO42	CL05B104KO5NNNC	10.050,2.600	90	C1525
C_IO49	CL05B104KO5NNNC	5.300,5.300	90	C1525
C_DV23	CL05B104KO5NNNC	6.600,-4.900	0	C1525
C_DV50	CL05B104KO5NNNC	4.200,5.300	90	C1525
C_USB	CL05B104KO5NNNC	6.100,7.400	90	C1525
C_ADC	CL05B104KO5NNNC	8.900,5.300	90	C1525
C_VREG_IN	CL05A105KA5NQNC	8.300,7.400	90	C52923
C_VREG_OUT	CL05A105KA5NQNC	7.200,7.400	90	C52923
U_FLASH	GD25Q16EEIGR	-4.500,6.100	0	C2986331
C_FLASH	CL05B104KO5NNNC	-7.000,8.000	0	C1525
R_FLASH_CS	0402WGF1002TCE	-1.800,7.900	90	C25744
R_BOOT	0402WGF1001TCE	14.700,4.000	0	C11702
Y_MCU	ABM8-272-T3	6.000,-7.300	0	C20625731
R_XOUT	0402WGF1001TCE	9.000,-6.600	90	C11702
C_XIN	0402CG150J500NT	3.200,-8.100	90	C1548
C_XOUT	0402CG150J500NT	9.000,-8.510	270	C1548
R_RUN	0402WGF1002TCE	-0.800,1.600	90	C25744
R_DM	0402WGF270JTCE	7.700,5.300	90	C25100
R_DP	0402WGF270JTCE	6.500,5.300	90	C25100
R_CC1	0603WAF5101T5E	15.900,6.500	90	C23186
R_CC2	0402WGF5101TCE	13.500,16.400	0	C25905
R_USB_SENSE_H	0603WAF1002T5E	12.100,3.700	0	C25804
R_USB_SENSE_L	0603WAF1002T5E	15.900,1.800	90	C25804

NEMA14_RP2040 - fitted placement reference
Native source convention; supplier pickup/rotation approval pending

Ref	Exact MPN	Native center mm	Rotation	JLCPCB
Q_DATA	MMBT3904-7-F	12.600,0.900	0	C94514
R_VM_H	0603WAF1003T5E	-15.900,16.400	0	C25803
R_VM_L	0603WAF1002T5E	14.700,-1.700	0	C25804
R_VM_BLEED	0603WAF4700T5E	10.800,-4.100	0	C23179
C_VM_SENSE	CL05B103KB5NNNC	13.700,-4.100	0	C15195
U_BUCK	AP63203WU-7	-5.600,-3.500	180	C780769
L_BUCK	SRN4018-4R7M	-7.800,-7.900	180	C780206
C_BOOTSTRAP	CL05B104KO5NNNC	-8.500,-2.600	90	C1525
C_BUCK_IN	CL32B106KBJNNNE	-6.400,0.150	0	C138687
C_BUCK_OUT1	CL21A226MAQNNNE	-7.900,-11.900	180	C45783
C_BUCK_OUT2	CL21A226MAQNNNE	-8.250,-16.000	180	C45783
C_BUCK_OUT3	CL21A226MAQNNNE	-8.300,-13.800	180	C45783
D_PD	B5819W SL	-4.000,2.900	0	C8598
D_DATA	B5819W SL	11.000,-7.700	90	C8598
U_DRV	A4988SETTR-T	-0.700,-8.850	270	C38437
C_CP	CC0603KRX7R9BB104	-4.650,-9.900	270	C14663
C_VCP	CC0603KRX7R9BB104	-4.900,-15.000	180	C14663
C_VREG	CL10B224KA8NNNC	-4.650,-12.850	90	C21120
C_DRV_LOGIC	CL05B104KO5NNNC	3.500,-11.550	0	C1525
C_VM	CC0603KRX7R9BB104	-4.650,-6.950	90	C14663
C_VM2	CC0603KRX7R9BB104	8.350,-10.160	180	C14663
C_BULK	EEEFPV101XAP	-13.750,-5.600	90	C178585
R_SA	FRL0805FR240TS	-0.300,-3.500	90	C2930216
R_SB	FRL0805FR240TS	-2.300,-3.500	90	C2930216
R_REF_H	0603WAF6201T5E	8.900,-12.800	90	C4260
R_REF_L	ERJPA3F1001V	5.100,-10.050	0	C441922
C_REF	CL10B103KB8NNNC	8.600,-15.600	0	C1589
R_ENABLE	0603WAF1002T5E	-2.300,-0.700	180	C25804
R_SLEEP	0603WAF1002T5E	0.000,13.000	0	C25804
J_MOTOR	B4B-PH-K-S(LF)(SN)	1.700,-14.100	0	C131334
U_ESD	USBLC6-2SC6	12.400,6.900	270	C7519
D_TVS	SMF4L16A	-13.200,1.800	0	C1973135